Aryan Achary

B.Tech Computer Science and Engineering (second year), Manipal Institute of Technology, India
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Interests: 3D vision (Gaussian splatting, multi-view segmentation); privacy and unlearning in scene representations.

Education

Qualification	Institution	Year	Score
B.Tech (CSE)	Manipal Institute of Technology, MAHE	2025–2029 (exp.)	CGPA 8.49 (2 sem.)
Class XII (CBSE)	Dalmia Vidya Mandir, Rajgangpur	2025	92%
Class X (CBSE)	Dalmia Vidya Mandir, Rajgangpur	2023	97.8%

Research

Auditing private-detail removal in 3D Gaussian Splatting Oct 2026
Research proposal, AAAI-27 Undergraduate Consortium (submitted, decision pending)

- Question: if a private detail is edited out of a 3DGS scene so that no render shows it, can it still be recovered from the Gaussian parameters? Edits are judged against a model retrained without the detail (the unlearning reference).
- Synthetic pilot with a random 256-bit pattern (9 runs per method): recolouring and fine-tuning made renders unreadable, but the pattern stayed readable from the parameters (bit accuracy 0.85–0.96 vs. 0.44–0.58 for the retrained model). Contribution-based selection, as used by common 2D-to-3D segmentation methods, missed about 3% of the Gaussians, which still carried it; rebuilding the whole 3D region reached the retrained range.
- Real phone capture of a desk, with the pattern inserted digitally onto a real blank sheet (2 seeds $\times$ 3 patterns): fine-tuning and the standard densify and opacity-reset clean-up left the pattern readable from per-cell Gaussian counts (0.84–1.00); only rebuilding the whole region stayed in the retrained model's range on all read-outs inside the region.

Object-landmark correction for dense 3DGS SLAM 2026
Equal-contribution co-author with Om Kumar; manuscript in preparation

- My part: lifting 2D object masks onto the Gaussian map, cleaning them with depth, visibility and multi-view checks, and the evaluation.
- On a dense 119-frame sequence, the cleaned landmark passed the correction guard in 20 of 24 trials (raw lift: 0 of 24) and cut trajectory error from 0.319 m to 0.144 m. With sparser frame sampling the correction raised error by 2–24%; the cause is still open.

Projects

Training-free distractor removal for 3D Gaussian Splatting 2026
Independent project

- Cleans an already-trained scene without retraining: SAM 2 propagates text-prompted masks through the capture, and each Gaussian is kept or pruned by a multi-view vote on where its centre projects, with a per-scene threshold.
- On 12 Nerfbusters scenes: 46–94% smaller files and higher held-out PSNR on 11 (largest gains 13.7 $\rightarrow$ 19.2 dB and 15.8 $\rightarrow$ 20.2 dB). The cleanup step takes 1.5–17 s and under 350 MB of memory per scene on a laptop.

Single-image height estimation from aerial imagery 2026
Smart India Hackathon 2026, ISRO/SAC problem statement, team project | github.com/Aryan241/Heimdall

- Trained a 1.72M-parameter decoder on top of a frozen Depth Anything V2 that predicts per-pixel scale and shift, turning relative depth into metric height above ground (16k GAMUS and Dutch LiDAR tiles). Chose the backbone by its correlation with LiDAR; a metric-depth model scored r = 0.06 on top-down imagery.
- On 8 held-out Dutch LiDAR sites: 5.32 m height RMSE (r = 0.67) vs. 8.37 m for predicting flat ground.

Leadership

- **Project MANAS**, student AI and robotics team, MIT Manipal: mentor 10+ junior members in machine learning and deep learning; run paper discussions and set coding exercises.
- **e-Yantra** robotics competition (IIT Bombay): team lead of a four-person team.

Skills

Languages: Python, C++, Java. **Libraries and tools:** PyTorch, gsplat, OpenSplat, COLMAP, SAM 2, Depth Anything, Transformers, OpenCV, rasterio, Git, Linux.